USN | | | | | | | | | | [23CSBSPH03]

# FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2024

## FUNDAMENTAL OF PHYSICS

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:**
- **i.** Missing data may be suitably assumed.
- **ii.** Physical constants
  Mass of the electron $m_e = 9.1 \times 10^{-31} kg$
  Boltzmann constant $k = 1.38 \times 10^{-23} J/K$
  Charge of the electron $e = 1.602 \times 10^{-19} C$
  Planck's constant $h = 6.626 \times 10^{-34} Js$
  Velocity of light $c = 3 \times 10^{8} m/s$
- **iii.** Answer all questions

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| | | **PART - A** | | | **05 X 02=10** |
| **1.** | **a** | Define Resonance and give an example. | L-1 | CO1 | **02** |
| | **b.** | Explain interference and diffraction of light. | L-2 | CO2 | **02** |
| | **c.** | What are matter waves and write the equation for de-broglie wavelength. | L-2 | CO4 | **02** |
| | **d.** | Mention any four properties of LASER. | L-2 | CO5 | **02** |
| | **e.** | State Zeroth law of thermodynamics. | L-4 | CO6 | **02** |
| | | **PART - B** | | | **06 X 05=30** |
| **2.** | **a.** | Describe Simple Harmonic Motion and mention any of its two characteristics. | L-2 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | What is meant by periodic motion and explain the concept of quality factor with an equation. | L-2 | CO1 | **05** |
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| **3.** | **a.** | Discuss any two types of interference. | L-2 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss the condition for circularly polarized light with a neat diagram. | L-2 | CO2 | **05** |
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| **4.** | **a.** | Mention two types of coherence and list the difference between them. | L-2 | CO3 | **05** |
| | | **OR** | | | |
| | **b.** | Explain Brewster's law and double refraction. | L-4 | CO4 | **05** |
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| **5.** | **a.** | Draw any five types of crystal systems. Mention lattice parameters for each. | L-2 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Explain Planck's quantum theory. | L-3 | CO5 | **05** |
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| **6.** | **a.** | Write a note on Einstein's coefficients. | L-4 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss any two Engineering applications of laser. | L-3 | CO6 | **05** |
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| **7.** | **a.** | Explain the concept of change of entropy in case of reversible process. | L-4 | CO2 | **05** |
| | | **OR** | | | |

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| | b. | Derive an expression for continuity equation for current density. | L-2 | CO4 | 05 |
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| | | **PART - C** | | **06 X010=60** | |
| 8. | a. | Explain the theory of forced mechanical oscillator A sonometer wire under tension is plucked and left free for oscillation. Find its frequency of vibrations if the midpoint on the string attains a maximum velocity of $1.57ms^{-1}$ when its amplitude of oscillation is 5 mm. Treat the vibration as simple harmonic (neglect the damping effect) | L-3 | CO1 | 10 |
| | | **OR** | | | |
| | b. | Discuss the theory of energy decay in a damped harmonic oscillator. | L-3 | CO1 | 10 |
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| 9. | a. | Describe the production of Polarized beam of light from two SHM acting at right angle with a neat diagram. | L-2 | CO2 | 10 |
| | | **OR** | | | |
| | b. | Classify the types of diffraction and explain the diffraction at single slit with a relevant fig and equations. | L-2 | CO2 | 10 |
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| 10. | a | Set up one dimensional time independent Schrodinger wave equation. Calculate the De-Broglie wavelength associated with an electron having a kinetic energy of 100eV. | 2 | CO3 | 10 |
| | | **OR** | | | |
| | b. | Define atomic packing factor and estimate the same for BCC and HCP structure with a neat diagram. | L-4 | CO4 | 10 |
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| 11. | a. | Describe the construction and working of Neodymium Laser with energy level diagram. | L-3 | CO5 | 10 |
| | | **OR** | | | 10 |
| | b. | Explain any two type of optical fibers with the relevant Refractive Index Profile and mode of propagation diagram. The N.A of an optical fiber is 0.2 when surrounded by air. Determine the refractive index of its core given the refractive index of the cladding is 1.59. Also find the acceptance angle when the fiber is in water. Assume the refractive index of water as 1.33. | L-2<br>L-3 | CO5<br>CO5 | 06<br>04 |
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| 12. | a. | Briefly discuss any one application of First and second law of thermodynamics. | L-2 | CO6 | 10 |
| | | **OR** | | | |
| | b. | List and explain the Maxwell's equations in vacuum and non-conducting medium. | L-2 | CO6 | 10 |
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| 13. | a. | Demonstrate the experimental determination of Planck's constant by studying LED's characteristics. | L-4 | CO1 | 10 |
| | | **OR** | | | |
| | b. | **i.** Describe the second law of thermodynamics and the concept of Engine.<br>**ii.** Briefly explain the third law of thermodynamics. | L-2<br>L-2 | CO5<br>CO5 | 06<br>04 |